

ASSIGNMENT No: 7

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TITLE: Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement: Implement a Java program to demonstrate inheritance and interfaces using suitable realworld examples.

Learning Objectives: To implement inheritance and interfaces for achieving code reusability and abstraction.

Theory:

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Exercise

1. Create a Shape application where Circle and Rectangle inherit from Shape and calculate area.

```
J ShapeDemo.java > Shape
1  class Shape {
2      void area() {
3          System.out.println(x: "Area Calculation");
4      }
5  }
6  class Circle extends Shape {
7      double radius = 5;
8
9      void area() {
10         double a = 3.14 * radius * radius;
11         System.out.println("Area of Circle = " + a);
12     }
13 }
14
15 class Rectangle extends Shape {
16     int length = 10, width = 5;
17
18     void area() {
19         int a = length * width;
20         System.out.println("Area of Rectangle = " + a);
21     }
22 }
23
24 public class ShapeDemo {
25     Run | Debug
26     public static void main(String[] args) {
27         Circle c = new Circle();
28         Rectangle r = new Rectangle();
29
30         c.area();
31         r.area();
32     }
33 }
```

PROBLEMS 6 OUTPUT DEBUG CONSOLE TERMINAL PORTS

[Done] exited with code=0 in 1.51 seconds

[Running] cd "c:\Users\arhaa\OneDrive\Desktop\java\" && javac
Area of Circle = 78.5
Area of Rectangle = 50

[Done] exited with code=0 in 0.896 seconds

2. Develop a E-Commerce Product System where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```
J Ecommerce.java > Grocery > Grocery(String, double)
1 interface Product {
2     void display();
3 }
4
5 class ProductDetails {
6     String name;
7     double price;
8
9     ProductDetails(String name, double price) {
10         this.name = name;
11         this.price = price;
12     }
13 }
14
15 class Electronic extends ProductDetails implements Product {
16     Electronic(String name, double price) {
17         super(name, price);
18     }
19
20     public void display() {
21         System.out.println("Electronic Product");
22         System.out.println("Name: " + name);
23         System.out.println("Price: " + price);
24     }
25 }
26
27 class Clothing extends ProductDetails implements Product {
28     Clothing(String name, double price) {
29         super(name, price);
30     }
31
32     public void display() {
33         System.out.println("Clothing Product");
34         System.out.println("Name: " + name);
35         System.out.println("Price: " + price);
36     }
37 }
38
39 class Grocery extends ProductDetails implements Product {
40     Grocery(String name, double price) {
41         super(name, price);
42     }
43
44     public void display() {
45         System.out.println("Grocery Product");
46         System.out.println("Name: " + name);
47         System.out.println("Price: " + price);
48     }
49 }
50
51 public class Ecommerce {
52     Run | Debug
53     public static void main(String[] args) {
54         Electronic e = new Electronic(name: "Laptop", price: 50000);
55         Clothing c = new Clothing(name: "T-Shirt", price: 800);
56         Grocery g = new Grocery(name: "Rice", price: 1200);
57
58         e.display();
59         System.out.println();
60         c.display();
61         System.out.println();
62         g.display();
63     }
64 }
65
66 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
$ C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Java\jdk-26.
g\Code\User\workspaceStorage\02e7f2cfce92a0ba8e30606ade6b2c24\redhat.java
Electronic Product
Name: Laptop
Price: 50000.0
Clothing Product
Name: T-Shirt
Price: 800.0
Grocery Product
Name: Rice
Price: 1200.0
$ C:\Users\arhaa\OneDrive\Desktop\java>
```